Build with AI

H2S

Solution Challenge

Team Details

a. **Team name:** FairLens AI Team
b. **Team leader name:** Vaishnavi Gaddam
c. **Problem Statement:** [Unbiased AI Decision] Ensuring Fairness and Detecting Bias in Automated Decisions - AI systems used in hiring, lending, and healthcare often encode historical biases, leading to discriminatory outcomes for marginalized groups.

FairLens AI is a real-time bias detection and fairness auditing platform powered by Google Gemini and Vertex AI. It automatically scans AI/ML models and datasets for demographic bias, generates explainable fairness reports, and recommends debiasing strategies - enabling organizations to build and deploy equitable AI systems across hiring, lending, healthcare, and public policy domains.

Opportunities

a. How different is it from any of the other existing ideas? Unlike IBM AI Fairness 360 or Google What-If Tool (which require data science expertise and manual setup), FairLens AI provides a no-code SaaS dashboard with automated bias scanning, plain-English audit reports, and Gemini-powered remediation suggestions - accessible to compliance officers, HR teams, and regulators with zero ML knowledge.

b. How will it be able to solve the problem? Users upload datasets or connect live model APIs; Vertex AI runs statistical fairness tests (disparate impact, equalized odds, demographic parity); Gemini explains findings in natural language; the platform auto-generates a compliance-ready PDF audit report and recommends reweighting or adversarial debiasing fixes.

c. USP of the proposed solution: First platform to combine automated multi-metric bias detection + Gemini-powered plain-English explanations + one-click debiasing recommendations in a compliance-ready dashboard. Supports models in any format (scikit-learn, TensorFlow, custom APIs) with no-code integration, making AI fairness auditing accessible to non-technical stakeholders.

Key Features of FairLens AI:

(1) Automated Bias Scanner - Upload CSV dataset or model endpoint; runs 8 fairness metrics automatically.

(2) Gemini Explainer - Translates statistical bias findings into plain-English audit reports with severity ratings.

(3) Debiasing Recommendations - Suggests reweighting, resampling, or adversarial debiasing strategies with code snippets.

(4) Multi-Domain Templates - Pre-built fairness checklists for hiring, credit scoring, healthcare triage, and criminal justice.

(5) Compliance Dashboard - Tracks fairness scores over time; generates PDF reports for EEOC, RBI, and EU AI Act compliance.

(6) API Integration - REST API and Python SDK to embed fairness checks directly into existing MLOps pipelines.

Process Flow / Use-Case Diagram:

USER uploads dataset or provides model API endpoint
→ INGESTION LAYER (Google Cloud Storage + Dataflow) validates and preprocesses data
→ BIAS ENGINE (Vertex AI) computes fairness metrics: Disparate Impact Ratio, Demographic Parity Difference, Equalized Odds, Predictive Parity
→ GEMINI ANALYSIS generates natural-language audit report with root cause analysis
→ RECOMMENDATION ENGINE suggests debiasing strategies with implementation code
→ COMPLIANCE DASHBOARD displays historical fairness trends and generates regulatory PDF reports
→ USER downloads report or triggers automated retraining pipeline.

Use Cases:

(A) HR team audits candidate screening model before deployment.
(B) Bank validates credit-scoring model for RBI compliance.
(C) Regulator performs third-party audit of a healthcare triage AI.

Wireframes / Mock Diagrams:

Screen 1 - UPLOAD PAGE: Drag-and-drop zone for CSV/model endpoint URL, protected attribute selector (gender, race, age), target variable selector, and domain template picker (Hiring / Credit / Healthcare).
Screen 2 - BIAS SCAN RESULTS: Color-coded metric cards (Green=Fair, Yellow=At-Risk, Red=Biased) showing Disparate Impact (0.74 - BIASED), Demographic Parity (0.12 - AT RISK), Equalized Odds (0.03 - FAIR). Gemini explanation panel below: "Your model approves loan applications for male applicants 35% more often than female applicants with identical financial profiles. This likely stems from historical gender imbalance in training data (1987-2010)."
Screen 3 - REMEDIATION PAGE: Side-by-side before/after fairness scores, recommended fix (Reweighing Algorithm), auto-generated Python code block, and one-click "Apply Fix and Retrain" button.
Screen 4 - COMPLIANCE REPORT: Downloadable PDF preview with audit timestamp, fairness scorecard table, and digital signature field.

Architecture Diagram of FairLens AI:

FRONTEND: React.js web dashboard + REST API (Cloud Run)
→ AUTHENTICATION: Firebase Auth (Google SSO)
→ DATA LAYER: Google Cloud Storage (dataset uploads) + BigQuery (audit logs and metrics history)
→ PROCESSING PIPELINE: Cloud Dataflow (data preprocessing and validation)
→ AI/ML CORE: Vertex AI Custom Jobs (bias metric computation - Disparate Impact, Equalized Odds, Demographic Parity, Calibration) + Vertex AI Model Registry (model versioning)
→ INTELLIGENCE LAYER: Gemini 1.5 Pro API (natural-language report generation and remediation advice)
→ RECOMMENDATION ENGINE: Custom Python service on Cloud Run (debiasing algorithm selector - Reweighing, Adversarial Debiasing, Calibrated Equalized Odds)
→ REPORTING: Cloud Functions trigger PDF generation (Puppeteer)
→ NOTIFICATION: Pub/Sub + SendGrid for compliance alerts
→ MONITORING: Cloud Monitoring + Looker Studio dashboards.

Technologies Used in FairLens AI:

GOOGLE AI SERVICES: Gemini 1.5 Pro (audit report generation and remediation advice), Vertex AI (custom bias computation jobs and model registry), Google Cloud Natural Language API (text analysis for unstructured data bias).

GOOGLE CLOUD INFRASTRUCTURE: Cloud Run (containerized API services), Cloud Storage (dataset storage), BigQuery (metrics warehouse), Cloud Dataflow (ETL pipeline), Cloud Functions (PDF generation triggers), Pub/Sub (event-driven alerts), Firebase Auth (authentication).

FRONTEND: React.js, Tailwind CSS, Chart.js (fairness metric visualizations).

BACKEND: Python (FastAPI), scikit-learn, AIF360 (IBM AI Fairness 360 library for metric computation), TensorFlow (adversarial debiasing). DEVOPS: Docker, GitHub Actions CI/CD, Artifact Registry.

REPORTING: Puppeteer (PDF generation), Looker Studio (compliance dashboards).

Estimated Implementation Cost (Monthly - MVP Scale): Google Cloud Run (API serving) - $15/month.
Cloud Storage (dataset storage, 100GB) - $2/month.
BigQuery (audit logs, 10GB processed) - $5/month. Vertex AI Custom Jobs (bias computation, ~50 audits/month) - $30/month.
Gemini 1.5 Pro API (report generation, ~50 audits x 5K tokens) - $20/month.
Firebase Auth (up to 10K users free) - $0/month.
Cloud Dataflow (data preprocessing) - $10/month.
Cloud Functions + Pub/Sub - $2/month.
TOTAL MVP COST: ~$84/month for up to 50 audits/month.
SCALE ESTIMATE: At 500 audits/month (enterprise tier) - ~$350/month.
Revenue model: Freemium (3 free audits/month) + Pro at $99/month (unlimited) + Enterprise at $999/month (API access + white-label reports).
Break-even at 9 Pro subscribers.

MVP Snapshots: The working prototype demonstrates:

(1) DATASET UPLOAD: User uploads the UCI Adult Income Dataset (48K rows); selects "sex" and "race" as protected attributes, "income" as target variable, and chooses the "Credit/Finance" compliance template.
(2) BIAS SCAN OUTPUT: Dashboard displays Disparate Impact Ratio = 0.36 (CRITICAL - far below 0.8 threshold), Demographic Parity Difference = 0.19 (HIGH RISK), Equalized Odds Difference = 0.08 (MODERATE). Gemini explanation: "The model predicts high income for white male applicants 2.8x more often than for Black female applicants with identical qualifications. Primary bias source: occupation and education features are proxies for race in pre-Civil Rights era training data."
(3) REMEDIATION: System recommends Reweighing Algorithm; displays before/after fairness scores; provides copy-paste Python code.
(4) COMPLIANCE REPORT: Auto-generated 4-page PDF with audit ID, metric table, root cause analysis, and recommended actions - ready for regulatory submission.

Future Development Roadmap:

PHASE 2 (Q3 2026) - Real-time Bias Monitoring: Continuous fairness monitoring for live production models via webhook integration; automated alerts when fairness scores degrade below thresholds; integration with MLflow and Kubeflow Pipelines for MLOps teams.
PHASE 3 (Q4 2026) - Multimodal Bias Detection: Extend beyond tabular data to detect bias in image recognition models (skin tone bias in medical imaging), NLP models (sentiment bias by gender/ethnicity in text classifiers), and video analysis systems.
PHASE 4 (2027) - Regulatory Intelligence: Auto-mapping of bias findings to specific clauses in EU AI Act, US Executive Order on Safe AI, and India DPDP Act; jurisdiction-specific compliance reports; integration with legal workflow tools. LONG-TERM VISION: Become the industry-standard fairness auditing layer embedded in every AI deployment pipeline globally - the "SonarQube for AI Ethics."

Provide links to your:

1. **GitHub Public Repository:** https://github.com/hiddentruth21/fairlens-prototype
2. **Demo Video Link (3 Minutes):** https://drive.google.com/file/d/19GW8bOE6CBqtidEvbFWfe07cngjMnLVg/view?usp=drive_link
3. **MVP Link:** https://fairlens-ai.web.app
4. **Working Prototype Link:** https://fairlens-ai.web.app/demo

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Thank You